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Made By Sharif Ashik Ishtiak.
Email: sharifashik591@gmail.com | WhatsAPP | Linkedin

AI-Powered Retail Intelligence Dashboard

Project Overview

This project presents an end-to-end Business Intelligence solution built using Microsoft Power BI. The goal of the project is to convert raw retail transaction data into meaningful insights that support strategic and operational decision-making in a retail environment.

In modern retail organizations, decision making is increasingly driven by data. Executives, sales managers, and operations teams need access to accurate performance indicators that summarize large volumes of transactional data. This dashboard was designed to provide a unified analytical view of business performance through interactive visualizations, business KPIs, and forecasting capabilities.

The solution connects to a PostgreSQL (PgSQL) database where the retail dataset is stored. The data is transformed and modeled using a Star Schema architecture before being visualized in Power BI.

The platform focuses on 4 major business objectives:

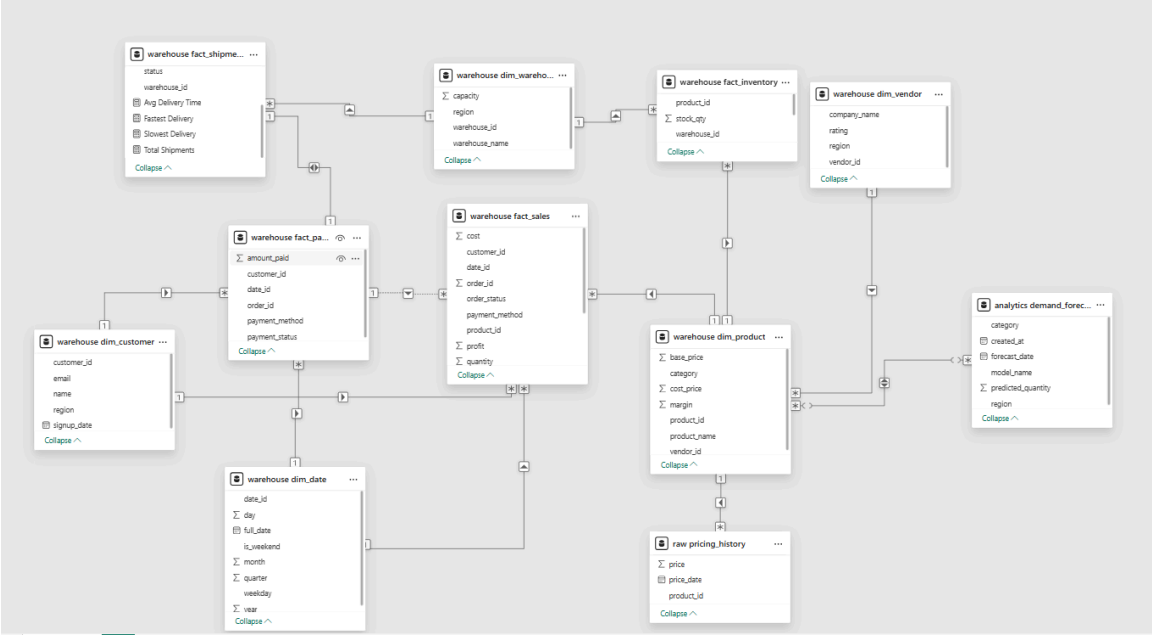
- Monitoring revenue and profitability
- Identifying product and category performance trends
- Supporting inventory and operational decision making
- Demand forecasting

Data Source and Data Architecture

The dashboard uses a hybrid data-engineering approach that brings together ERP data from a MySQL source and a centralized PostgreSQL data warehouse. The Power BI dashboard connects directly to the PostgreSQL database using a secure data connection. This enables automated refresh capabilities and ensures that the dashboard always reflects the most recent data available.

After connecting to the database, the data was cleaned, transformed, and structured using Power Query before creating the analytical model used in the dashboard.

Made By Sharif Ashik Ishtiak.
Email: sharifashik591@gmail.com | WhatsAPP | Linkedin



Data Modeling Approach – Star Schema

A Star Schema data model was implemented in this project to ensure efficient querying and optimal performance in Power BI. In a Star Schema structure, the data model consists of one central Fact Table surrounded by multiple Dimension Tables. The fact table contains measurable business metrics such as sales, revenue, and profit, while dimension tables provide descriptive context such as product information, categories, and dates.

Typical structure used in this project:

Fact Table:

- Sales Transactions

Dimension Tables:

- Product Dimension
- Category Dimension
- Date Dimension
- Customer or Order Dimension (if applicable)

This structure improves analytical performance because Power BI can efficiently aggregate the fact table while using dimension tables to filter and segment the data.

Made By Sharif Ashik Ishtiak.
Email: sharifashik591@gmail.com | WhatsAPP | Linkedin

Dashboard Page 1 – Executive Business Overview

Purpose of the Page:

The Executive Overview page provides a high-level summary of overall business performance. This page is designed specifically for business leaders and decision makers who need a quick snapshot of the organization's performance.

Key Metrics Included:

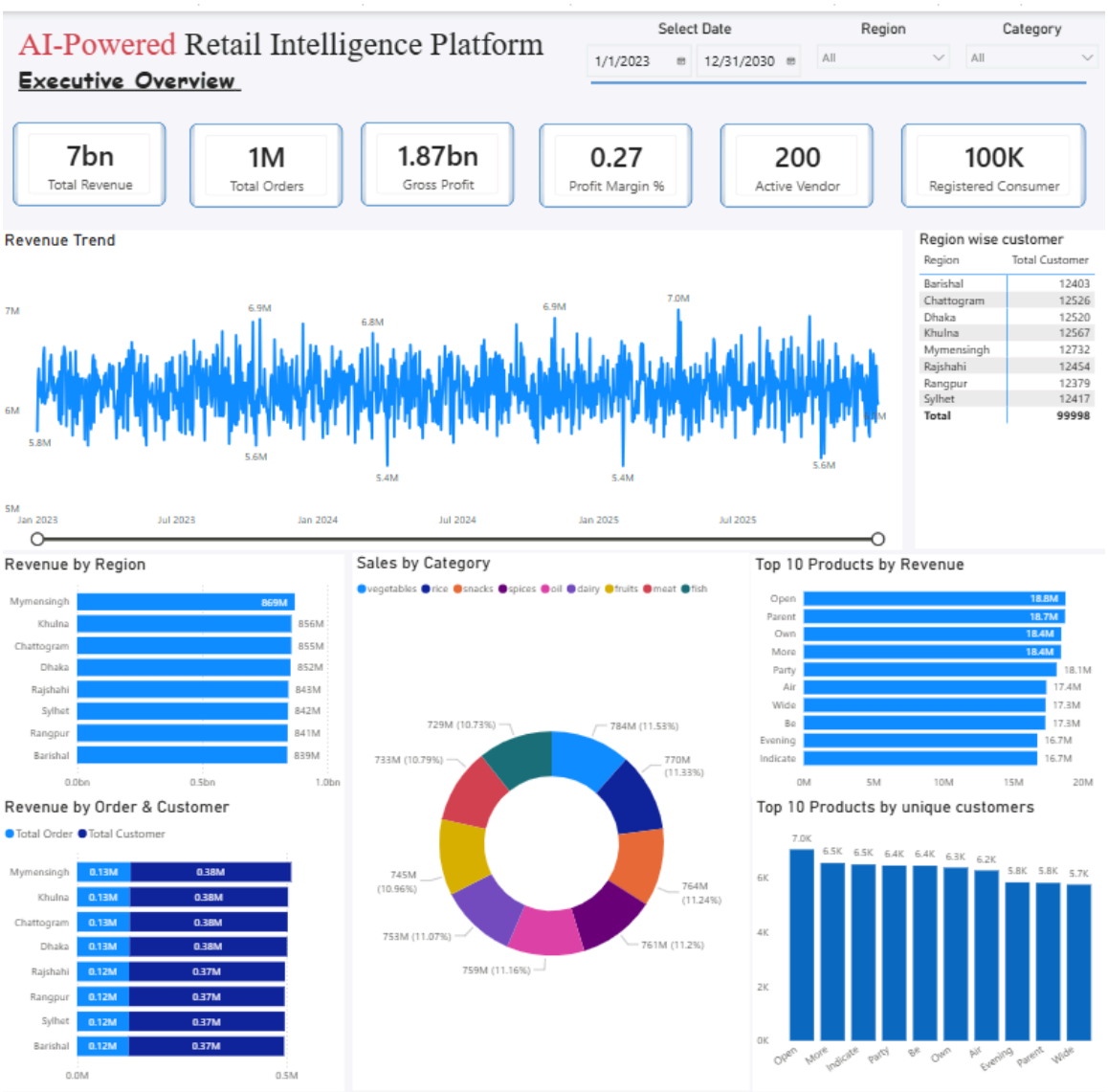
- Total Sales
- Total Orders
- Total Profit
- Sales Trends Over Time
- Category-level Sales Distribution

Business Value:

This page allows executives to quickly evaluate whether the business is meeting its financial goals. For example, Total Sales provides a direct view of revenue generation, while Total Profit indicates the sustainability and efficiency of operations.

Sales trend analysis allows leadership to identify growth patterns or sudden declines in revenue. If sales are decreasing during a specific period, executives can investigate whether the issue is related to seasonality, supply shortages, pricing strategies, or competitive pressure.

The category distribution visualization helps decision makers identify which product categories are driving the majority of revenue. This insight can guide strategic decisions such as marketing investments, inventory allocation, and promotional campaigns.



Dashboard Page 2 – Sales Forecasting & Product Performance

Purpose of the Page:

This page focuses on deeper analytical insights, particularly around product performance and future demand forecasting.

- Key Metrics and Visualizations:
- Historical Sales Trend
- Forecasted Sales
- Top Performing Products
- Product Revenue Contribution
- Category-wise Product Performance

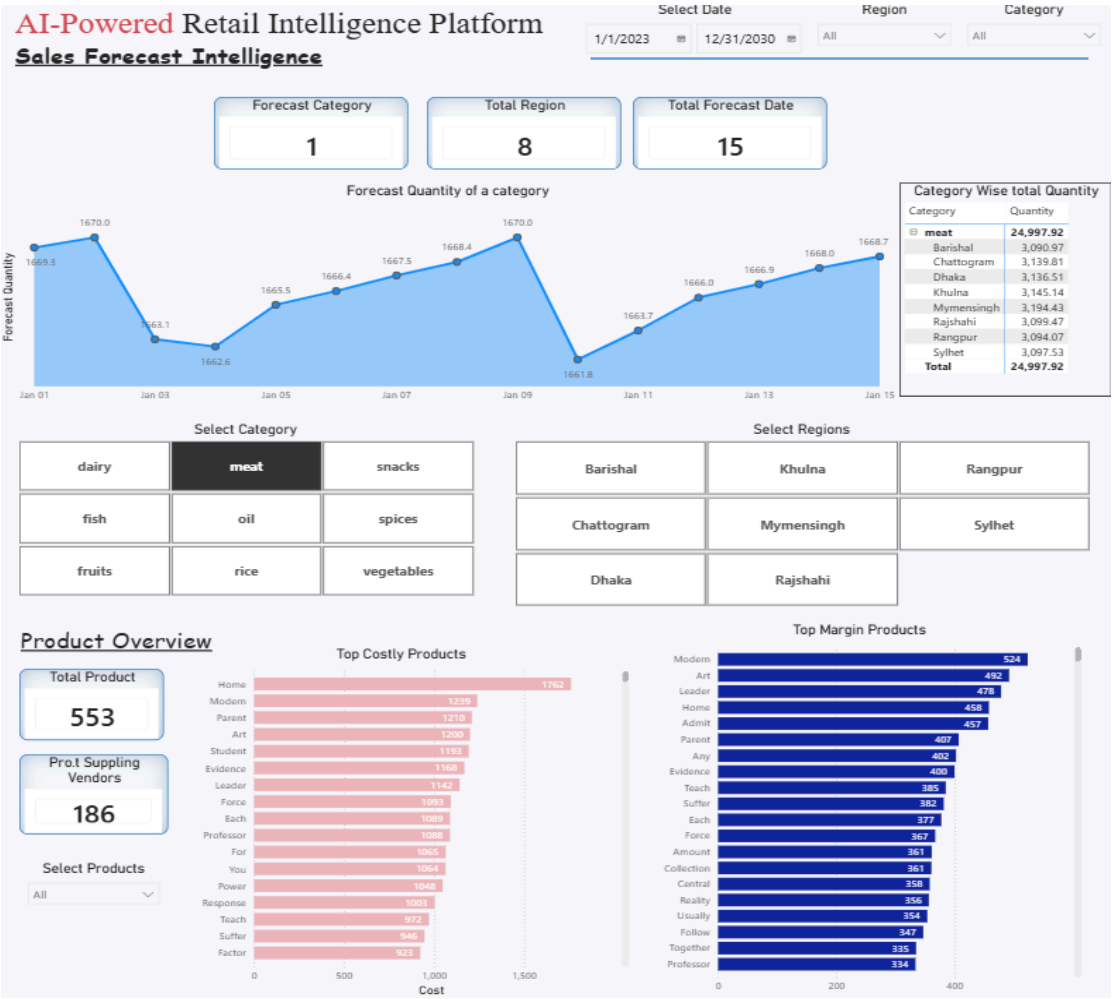
Business Value:

Retail businesses must continuously anticipate future demand to maintain operational efficiency. Sales forecasting plays a critical role in planning procurement, staffing, and logistics.

By analyzing historical sales patterns, the forecasting model estimates future revenue trends. This allows businesses to proactively prepare for periods of high demand and prevent stock shortages.

Product performance metrics highlight which items generate the highest revenue. Businesses can use this information to prioritize marketing campaigns, negotiate supplier contracts, and optimize product placement in stores or online platforms.

For example, if a small number of products contribute a large percentage of total sales, the company may decide to increase inventory levels for those products to prevent lost sales.



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Dashboard Page 3 – Operational & Inventory Insights

Purpose of the Page:

This page supports operational teams responsible for inventory management and supply chain operations.

Key Metrics:

- Inventory Levels
- Product Turnover
- Product Availability
- Inventory Risk Indicators

Business Value:

Inventory management is one of the most critical components of retail operations. Maintaining too much inventory increases storage costs, while insufficient inventory leads to lost sales opportunities.

This dashboard page enables operations teams to monitor product movement and identify slow-moving or fast-moving items.

High product turnover indicates strong demand, suggesting that the business should increase procurement levels. On the other hand, slow-moving inventory may indicate pricing issues, declining customer demand, or overstocking.

Using these insights, managers can adjust purchasing strategies, optimize warehouse capacity, and reduce operational costs.

Business Impact of the Dashboard

This AI-Powered Retail Intelligence dashboard enables organizations to transition from reactive decision-making to proactive and data-driven strategies.

Key organizational benefits include:

- Improved revenue monitoring
- Better demand forecasting
- Optimized inventory management
- Faster executive decision making
- Enhanced operational efficiency

By integrating data modeling, business KPIs, and predictive analytics within a single platform, the solution provides a comprehensive view of the retail business ecosystem.

Made By Sharif Ashik Ishtiak.
Email: sharifashik591@gmail.com | WhatsAPP | Linkedin